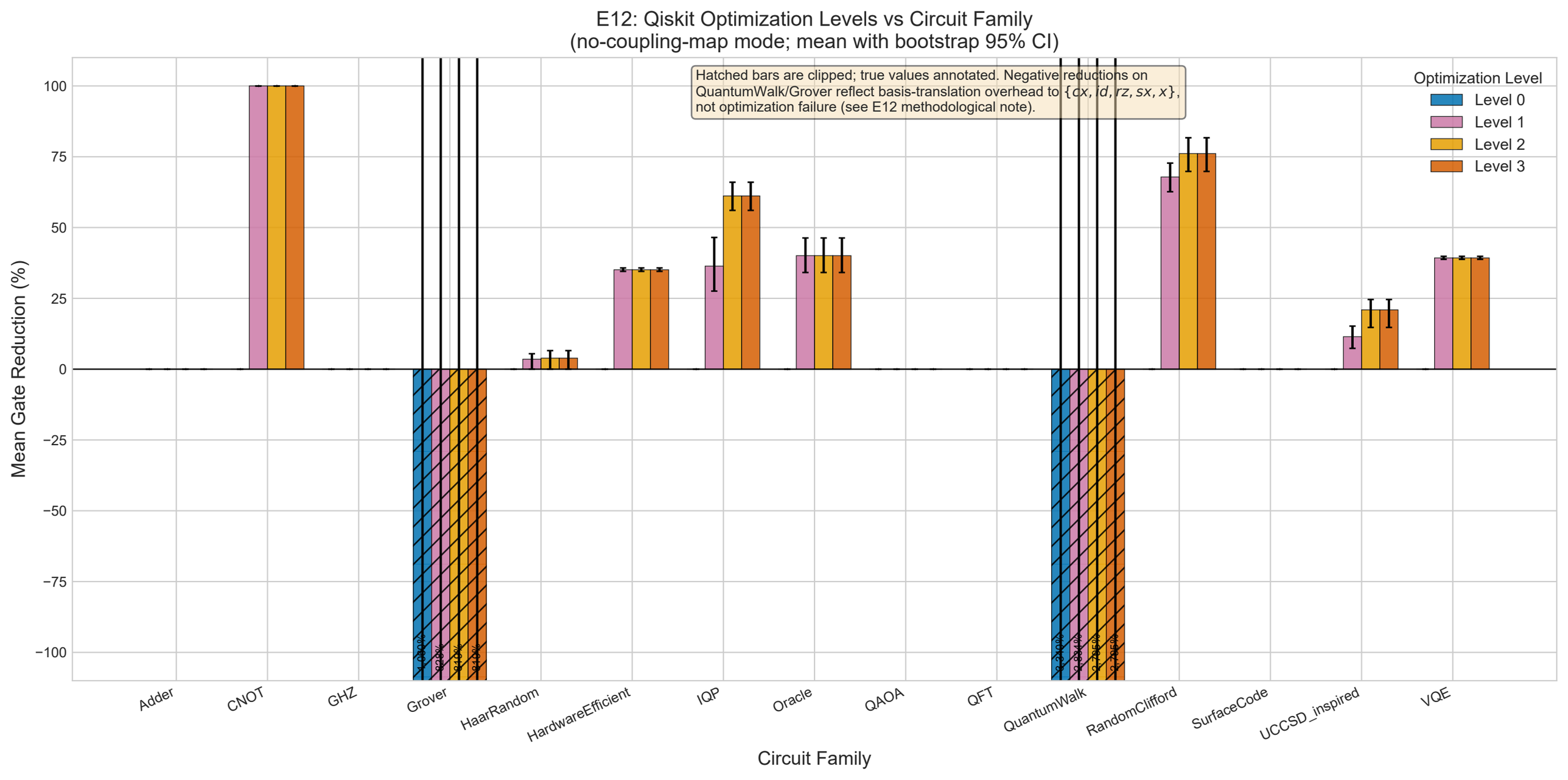
E12: Qiskit Optimization Levels vs Circuit Family
(no-coupling-map mode; mean with bootstrap 95% CI)
Hatched bars are clipped; true values annotated. Negative reductions on QuantumWalk/Grover reflect basis-translation overhead to {cx, id, rz, sx, x}, not optimization failure (see E12 methodological note).
Optimization Level
Level 0
Level 1
Level 2
Level 3
Mean Gate Reduction (%)
100
75
50
25
0
−25
−50
−75
−100
-1.000%
-923%
-819%
-819%
-2.304%
-2.304%
-2.705%
-2.705%
Adder
CNOT
GHZ
Grover
HaarRandom
HardwareEfficient
IQP
Oracle
QAOA
QFT
QuantumWalk
RandomClifford
SurfaceCode
UCCSD_inspired
VQE
Circuit Family